

PATH 548L Quiz 3 2025W2 May 2028, 2026

Quiz rules: There are 24 marks possible.

You can work in teams of 2-3 if you wish and submit a single set of answers with all team member names. Answers to quiz questions must be in narrative, tabular or graphical format as appropriate. We expect narrative answers to be 3-7 sentences maximum!

We want to know the rationale of why you have chosen to address the problem in the way that you have. Although JASP and Python are used in the course, you are free to use any software that you like to answer the questions.

The quiz is **take-home** and consequently open-book with internet access.

Students are permitted to use artificial intelligence tools, including generative AI, to gather information, review concepts or to help produce assignments. However, students are ultimately accountable for the work they submit, and any content generated or supported by an artificial intelligence tool must be cited appropriately. Please be aware of what constitutes academic misconduct at UBC. You can find information here <https://academicintegrity.ubc.ca/regulation-process/academic-misconduct/>.

The JASP info buttons could be an extremely valuable resource for you. If you have difficulty in JASP or python, you can communicate with Andy or Helene on piazza or by email (aroeth@bccrc.ca or Helene.cote@ubc.ca).

Please turn the quiz in as a PDF document in the form **lastname_quiz3.pdf** i.e. **roth_quiz3.pdf**. If submitting as a team, include all last names in alphabetical order separated by an underscore i.e. **cote_roth_quiz3.pdf**. The quiz is due May 28 @1pm by email to both of us (aroeth@bccrc.ca; Helene.cote@ubc.ca)

Please advise Andy and Helene if you experience circumstances that will prevent you from making this deadline.

QUIZ START

Question 1 (4 marks): State the assumptions for multivariable linear regression and briefly explain how to test for them.

Question 2 (4 marks): Using the Follicular Lymphoma dataset fl.xlsx, perform a set of univariable survival analyses to determine whether any of the biomarkers: CD8_LAG3grp, FLIPIgrp, or M7Fgrp are significantly prognostic. Present your results in a graphical format fit for publication with an informative figure legend.

Question 3 (6 marks): Continuing on with the Follicular Lymphoma data and the biomarkers that showed prognostic significance from Question 2, perform a multivariable survival analysis that controls for: AGE, SEX, TREATMENT, and DIAG. Using the method of your choice, eliminate variables that are unnecessary from the model and report the Risk (Hazard) Ratios for the biomarkers to determine if they are independently prognostic and present these findings in a publication quality table or figure. Refer to the Data Dictionary for more information on these variables.

Question 4 (2 marks): We suggested in class that creating categories from continuous variables is a bad idea. CD8_LAG3grp, FLIPIgrp and M7Fgrp are actually categorised versions of the CD8_LAG3, FLIPI and M7FLIPI variables. Use your model from question 3, and replace the categorised variables with their continuous equivalent. How do your results change?

Question 5 (8 marks): Using the dataset body_fat.xlsx, perform a multivariable analysis to determine the set of body measurements and demographics that best predicts percent body fat. Ensure to check your assumptions that you identified in Question 1. Provide a report on the model fit, parameter estimates and 95% confidence intervals. Present these findings in a publication quality table or figure. Do you believe your approach is the best way to model the data? Why or why not?

- ***Be sure to clearly state the statistical methodology and any assumption checks used.***
- ***Ensure all figures and tables are of publication quality***
- ***Ensure your report is logically structured and easy to follow***